

Performance Evaluation of E-Commerce Operations in Excel

[Prepared by- MD. PARVES ALI | m.parves1998@gmail.com]

Introduction:

In today's fast-paced digital world, e-commerce businesses generate vast amounts of data. However, raw data alone provides little value without proper analysis and data visualization. This project aims to address this challenge by demonstrating how to leverage Microsoft Excel—a widely accessible and powerful tool—to create dynamic and insightful performance evaluations of e-commerce operations. By using Excel **functions, Charts, Pivot Tables, IF(S), Lookup (V, X, H), Index-Matching, Statistical Analysis, Forecast Sheet, Macro & VBA, conditional formatting, and finally dashboard** businesses can gain meaningful insights into their operational performance and identify areas for improvement. The objective is to demonstrate how Excel can serve as a practical tool for transforming raw e-commerce data into actionable business intelligence.

Objective:

Microsoft Excel is the industry-leading spreadsheet software program, a powerful data visualization and analysis tool. After learning Excel, I aim to gain practical experience with Microsoft Excel to showcase my Excel skills and keep my technical skills updated for today's demands. I aim to analyze **Sales and Revenue Trends performance** evaluation of E-commerce operations in Excel.

Data Collection:

I have generated the raw data for this project using AI. This data is exclusively intended for learning-based project work and serves as a means to effectively showcase my Microsoft Excel skills. The sample raw data provided below illustrates the type of information that will be utilized for in-depth analysis and dashboard creation.

Data Analysis:

After collecting the **raw data**, I cleaned it by handling missing values, standardizing the format, and organizing it into a table named “Clean Data Table.” Now, let’s take a look at the overall performance of this e-commerce operation.

Sum of Sales	Sum of Profit
8120	1865

Country	Sum of Sales	Sum of Profit
Australia	645	180
Canada	2865	670
UK	1250	301
USA	3360	714
Grand Total	8120	1865

Using **Pivot Tables**, I analyzed the profit, which totals \$1,865. The majority of this profit comes from the USA and Canada, contributing \$714 and \$670, respectively. These two countries also ranked as the top two sellers during the period analyzed. Overall, the organization is on the right track, with a profit margin of approximately 23%.

Additionally, using **VLOOKUP** function, I solve this question- “Analyze the Sales data. Assuming there's a tiered discount structure where customers get 5% off for total sales between \$100-\$499, 10% for \$500-\$999, and 15% for \$1000+, use VLOOKUP and ROI with an approximate match to determine the applicable discount percentage for each Order ID's total Sales. Then, calculate the adjusted Revenue after applying this discount”. I also determine ROI in this question. Here is the sample of my answer-

Sales Min	Discount %
0	0%
100	5%
500	10%
1000	15%

This data provides a daily snapshot of sales performance, including Total Sale, Discount % To Be applied, calculated Discount Value, resulting Revenue, and ROI (%). We can observe that discounts are not applied uniformly; some sales have a 0% discount, while others range up to 15%. For instance, on 1/5/2022, a 15% discount on a 1200-unit sale resulted in a significant 180 discount value and 1020 in revenue, with an ROI of 85%. In contrast, many entries show a 0% discount and consistently achieve a 100% ROI, suggesting that price sales are highly profitable. The ROI (%) frequently stands at either 85% or 100%, indicating a consistent

profitability model tied to whether a discount is applied. The data spans from January 2022 to January 2024, allowing for an analysis of trends in sales volume, discount application, and their impact on revenue and profitability over time. I also solved various problems using XLOOKUP, HLOOKUP and INDEX-MATCH, and the question are given below-

Column 1	Total Sale	Discount % To Be	Discount Value	Revenue	ROI (%)
1/5/2022	1200	15%	180	1020	85
1/5/2022	25	0%	0	25	100
1/6/2022	150	5%	7.5	142.5	95
1/7/2022	30	0%	0	30	100
1/8/2022	60	0%	0	60	100
2/8/2022	75	0%	0	75	100
2/8/2022	100	5%	5	95	95
3/1/2022	300	5%	15	285	95
3/5/2022	45	0%	0	45	100
4/6/2022	80	0%	0	80	100
5/8/2022	10	0%	0	10	100
6/1/2022	250	5%	12.5	237.5	95
7/5/2022	120	5%	6	114	95
8/7/2022	70	0%	0	70	100
9/9/2022	90	0%	0	90	100
8/9/2022	80	0%	0	80	100
1/8/2022	35	0%	0	35	100
8/5/2022	50	0%	0	50	100
1/5/2023	400	5%	20	380	95
1/9/2023	95	0%	0	95	100
2/9/2023	800	10%	80	720	90
3/8/2023	120	5%	6	114	95
4/7/2023	70	0%	0	70	100
5/1/2023	100	5%	5	95	95
6/5/2023	40	0%	0	40	100
7/5/2023	90	0%	0	90	100
8/5/2023	180	5%	9	171	95
9/2/2023	150	5%	7.5	142.5	95
1/2/2023	350	5%	17.5	332.5	95
1/1/2023	110	5%	5.5	104.5	95
1/5/2023	50	0%	0	50	100
1/5/2024	1300	15%	195	1105	85
1/1/2024	180	5%	9	171	95
2/1/2024	150	5%	7.5	142.5	95
3/2/2024	130	5%	6.5	123.5	95
4/2/2024	70	0%	0	70	100
5/1/2024	100	5%	5	95	95
6/1/2024	200	5%	10	190	95
7/1/2024	60	0%	0	60	100
8/2/2024	80	0%	0	80	100
9/2/2024	20	0%	0	20	100
9/3/2024	450	5%	22.5	427.5	95
1/5/2024	25	0%	0	25	100

- Imagine you have a separate summary table where Campaign Names are listed vertically, and target metrics for different quarters are listed horizontally. Use **HLOOKUP** to dynamically retrieve the target Leads for a specific Campaign channel for 2022-2024.
- Find the Spend (use **INDEX-MATCH**) of the Marketing Campaign order to spend.
- Find the Customer ID using **XLOOKUP**.

In addition, I performed a **statistical analysis** of this e-commerce data. Below is the overall summary, and for better understanding, Excel is an effective tool to use. I also incorporated a Slicer to make it easier to filter and observe specific data points interactively. Here's a data analysis of the provided profit data, incorporating the summary statistics, monthly breakdown, and histogram:

Profit Analy▼	Column1▼
Elements	Numbers
Mean	37.79069767
Standard Error	6.845030219
Median	25
Mode	15
Standard Devi	44.88586486
Sample Varian	2014.740864
Kurtosis	14.92602133
Skewness	3.516703537
Range	257
Minimum	3
Maximum	260
Sum	1625
Count	43
Confidence Le	13.81383024



Column Labels▼	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	#VALUE!
Sum of Profit	353	365	120	260	85	68	53	115	77	83	90	176	20
Count of Profit	7	4	4	3	3	3	3	3	3	3	3	3	2
Average of Profit	50.42857143	91.25	30	86.66666667	28.33333333	22.66666667	17.66666667	38.33333333	25.66666667	27.66666667	30	58.66666667	10
Max of Profit	240	260	40	160	50	28	30	50	45	40	60	90	10
Min of Profit	8	5	15	40	10	15	3	15	12	18	8	16	10
StdDev of Profit	84.19987558	117.0736947	10.8012345	64.29100507	20.20725942	6.806859286	13.65039682	20.20725942	17.21433511	11.2398102	26.90724809	38.27967259	0
Var of Profit	7089.619048	13706.25	116.6666667	4133.333333	408.3333333	46.33333333	186.3333333	408.3333333	296.3333333	126.3333333	724	1465.333333	0

Country

Australia

Canada

UK

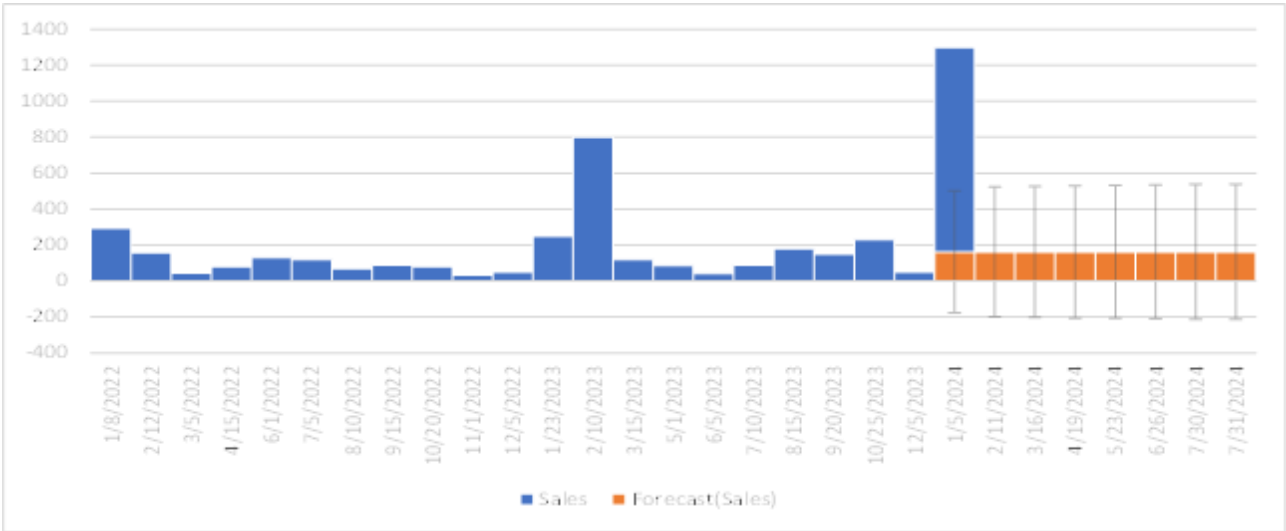
USA

This set of data provides a comprehensive "Profit Analysis" across different dimensions. The descriptive statistics indicate that the average profit is approximately 37.79, but with a high standard deviation of 44.89, suggesting a wide spread in profit figures. The median profit is 25, and the mode is 15, both significantly lower than the mean, which, coupled with a high positive skewness of 3.52, suggests the profit distribution is heavily skewed towards lower values, with a few exceptionally high profit instances pulling the mean up. This is further supported by the histogram, which visually confirms a large concentration of profit instances falling within the lower ranges (specifically around the 60 mark or less), with very few instances of higher profits (e.g., around 180 and 270).

The monthly profit breakdown reveals significant fluctuations throughout the year. January (353) and February (365) show the highest total profits, while July (53) and June (68) exhibit the lowest total profits, indicating a potential seasonal trend in profitability. The average profit also varies considerably month-to-month, with February having the highest average profit of 91.25 and July

having the lowest at 17.67. This detailed monthly view highlights periods of strong and weak performance, which could be crucial for operational planning and identifying underlying causes for profit variances.

E-commerce forecasting involves analyzing past sales data, customer behavior, and market trends to estimate future product demand. It's a crucial process for online retailers to make informed decisions about inventory management, marketing campaigns, and overall business strategy. This bar chart illustrates historical sales performance alongside a future **sales forecast**, covering a period from January 2022 through July 2024. The blue bars represent actual sales data, showing considerable fluctuation, with notable peaks in sales around 1/8/2022 (approx. 290 units), 2/10/2023 (approx. 790 units), and a significant spike on 1/5/2024 (exceeding 1200 units). There are also periods of relatively low sales, such as in March 2022, November 2022, and January 2023.



The orange bars, starting from February 2024 and extending to July 2024, represent the forecasted sales, which appear to be consistently low (around 160 units) with substantial error bars. The large error bars indicate a high degree of uncertainty or variability in these future sales predictions. This forecast suggests a sharp decline in sales after the peak in January 2024 and a sustained period of lower sales, highlighting a potential area for concern or further investigation into the factors driving the projected downturn.

The ecommerce industry is experiencing rapid growth, and it's expected to reach a staggering \$6.8 trillion by 2028. This figure highlights the transformation of the global retail landscape, where more people are opting for online shopping rather than visiting brick-and-mortar stores. In 2025, the global ecommerce market was already valued at over \$5 trillion, and by 2028, it's expected to reach \$6.8 trillion. This tremendous growth is not just a trend; it's a fundamental shift in how people approach shopping.

Finally, **Excel Dashboards** include various elements such as charts, tables, figures, and gauges that help in presenting the data in very short time. This Excel Dashboard provides a comprehensive overview of your E-Commerce operations, offering valuable insights into sales performance, profitability, product distribution, customer activity, and geographical reach.

Broad Analysis of the E-Commerce Sales Dashboard:

1. Executive Summary Panel (Top Section):

- **Total Sales:** 8120: This is your primary revenue figure. It's a good starting point, but its significance depends on the period it represents (e.g., year-to-date, specific year).
- **Total Profit:** 1865: This indicates the overall profit generated from the sales.
- **Profit (%):** 22.9%: This is a healthy profit margin, suggesting good cost management relative to sales.
- **Profit Forecasting:** 4.35%: This number is a bit ambiguous without context (e.g., is it a growth rate, a forecast margin?). Assuming it's a forecasted profit growth, it's a modest but positive outlook.



2. Sales & Profit Trends Over Time:

Sales (\$) and Profit (%) by Month (Top Right Bar Chart):

- **Sales Dominance in January:** There's a very significant spike in sales in January (around 3800 units), with other months being considerably lower, generally staying below 1000 units. This points to a strong seasonal pattern, likely influenced by new year sales, holiday hangover purchases, or specific promotional events in January.

- **Profit % Fluctuation:** The profit percentage (small lines on the bars) also varies by month, but broadly mirrors the sales volume.

Monthly Profit (Bottom Right Line Chart):

This chart provides a more granular view of profit trends over time (2022-2024).

- **2022:** Profit was relatively low and stable in the first half, with a slight peak mid-year.
- **2023:** Shows a more dynamic pattern with higher profits in early 2023 (consistent with the January sales peak) and a peak around July/August, followed by a decline towards the end of the year.
- **2024 (Partial):** Starts strong in January (again, consistent with the January sales peak), then seems to drop.

3. Product Performance:

Products Sales Per Year (Doughnut Chart):

- Electronics (1590) dominates product sales, accounting for 59% (calculated: $1590 / (260+560+1590) = \sim 59\%$). This is the core revenue driver.
- Home Goods (560) is the second largest at 21%.
- Apparel (260) and Footwear (240 - implicitly from the percentage chart) are smaller categories.

The total displayed for 2022, 2023, 2024 match the dashboard's "Total Sales: 8120," indicating that the doughnut chart likely represents three years sales.

Products Percentage (Bottom Left Bar Chart):

- Electronics: 28.0%: This percentage is significantly lower than the 59% implied by the doughnut chart's 2023 sales. This discrepancy needs to be clarified. It might be representing a different metric (e.g., percentage of items sold, not revenue) or a different time period.
- Apparel: 11%
- Home Goods: 11%
- Footwear: 10%

Many other product show very low percentages (below 2%), indicating a long tail of less frequently sold items.

4. Top Selling Product:

Electronics: \$5,820: This explicitly confirms Electronics as your primary revenue generator. The value (\$5,820) is much higher than the \$1,590 shown in the doughnut chart, reinforcing the idea that the doughnut chart is only for 2023, and Electronics has been performing strongly across the entire period covered by the dashboard's total sales.

5. Customer Performance:

Sales by Customer (Middle Left Bar Chart):

- Customer C001 is your absolute top customer, contributing over 2,500 in sales. This customer is immensely valuable.
- C005 and C002 also show significant sales.
- Many customers contribute very little, following a typical Pareto principle (80/20 rule).

6. Geographical Sales Distribution:

Sales Percentage by Country (Middle Right Doughnut Chart):

- Australia (40%) and UK (26%) are your primary markets, together accounting for 66% of sales.
- USA (16%) is a notable third.
- Canada (18%) is also a significant market.
- UK represents smaller markets.

Recommendations

Based on the analysis, here are some of my recommendations for E-Commerce operations:

- Investigate what drives the mid-year profit peaks (e.g., July/August 2023). Are these specific campaigns, product launches, or external factors?
- Electronics Focus: Continue to prioritize "Electronics" given its status as the top-selling product and major revenue contributor. Ensure stock levels are robust, marketing efforts are strong, and competitive pricing is maintained.
- While many products contribute little individually, collectively they can add up. Consider strategies for lower-selling items:
- Bundling: Group low-performing items with popular ones. Run specific campaigns for less popular categories.
- Nurture Top Customers: Customer C001 is incredibly valuable. Implement a loyalty program, offer personalized promotions, or provide exclusive benefits to retain and grow sales from your top customers (C001, C005, C002).
- Prioritize UK & Australia: Given their significant contribution, focus marketing and logistics efforts on these regions.
- Profit Margin Analysis: While 22.9% is good, continuous monitoring of profit margins per product and per sale is crucial. Identify any product categories or sales periods where margins are unusually low and investigate the reasons (e.g., high returns, increased costs, aggressive discounting).
- Inventory Management: The sales fluctuations suggest a need for dynamic inventory management to avoid overstocking during low periods and understocking during peaks, especially for popular Electronics items.

Conclusion:

This e-commerce performance evaluation, driven by comprehensive Excel analysis, reveals a dynamic operational landscape. The business maintains a healthy overall profit margin of 22.9%, with total sales reaching 8120 and profit at 1865. A pronounced seasonal sales pattern is evident, with significant peaks in January, strongly driven by Electronics, which stands out as the top-selling product category. While Australia and the UK are the dominant sales territories, and Customer C001 is a critically valuable asset, the data also highlights the need for consistent data representation across dashboards and a deeper understanding of forecasted trends. Leveraging these insights through targeted strategies for seasonal marketing, product optimization, customer relationship management, and geographical focus will be key to sustaining and enhancing the e-commerce operation's growth and profitability.

Excel Link:

<https://drive.google.com/drive/u/0/folders/1I5HxI9AlCwclYQU3mpTyfuqJRMESg2tv>